Geometry Unit 7 Assessment Guide

General Information	Unit 7 introduces students to trigonometric ratios and special right triangles. Students learn the Pythagorean theorem in Grade 8 and apply it again here, in unit 7. Question 3 may provide teachers with feedback on which relationships within a right triangle a student has not yet connected with. It is suggested that teachers monitor which, if any, answers a majority of students miss. This may offer the teacher insight on which relationships students may not have grasped. Monitor how well students do on question 5. Students may struggle on diagrams with overlapping triangles. Many of the questions have multiple approaches. Allow students the ability to choose an approach that works for them. On occasion a student may accidently use radian mode on their calculator. Teachers should consider how much this error should affect a student's grade. If a student is showing knowledge of the benchmark but makes this error, teachers should consider an overall deduction that emphasizes the necessary preciseness while acknowledging that a student is showing understanding of the content. Also, students are asked to respond with their answers in a variety of formats. Although the preciseness of a student response is important, teachers are also trying to determine what content the student has learned. Teachers should consider such errors as minor and deduct a small number of points. The draft test item specifications state that students will not be assessed on connections to the unit circle.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.T.1.2
6.928 feet		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**2a**	**Benchmark(s)**	MA.912.T.1.1
$\frac{38.4}{41.6}$		Recommended Scoring (7 Total Points)	All equivalent values/ratios should be accepted.
Question	**2b**	**Benchmark(s)**	MA.912.T.1.1
$\frac{8}{19.2}$		Recommended Scoring (7 Total Points)	All equivalent values/ratios should be accepted.
Question	**2c**	**Benchmark(s)**	MA.912.T.1.1
By using angle K as the reference angle, it can be shown that the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ for ΔHMK [○ is / ○ is not] equivalent to the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ for ΔRBK. [○ All / ○ No] similar triangles that use angle K as the reference angle [○ will / ○ will not] have an equivalent $\frac{\text{opposite}}{\text{hypotenuse}}$ ratio.		Recommended Scoring (7 Total Points)	Consider no partial credit. Students should complete both statements correctly to demonstrate understanding.

Question	3	Benchmark(s)	MA.912.T.1.2
☐ $\tan(x) = \frac{9}{w}$ ☐ $9^2 + w^2 = \left(\sqrt{125}\right)^2$ ☐ $\cos(y) = \frac{w}{\sqrt{125}}$ ☐ $w = \sqrt{9^2 + \left(\sqrt{125}\right)^2}$ ☐ $\sin(90 - x) = \frac{9}{w}$ ☐ $\sin(36.4°) = \frac{w}{\sqrt{125}}$ ☐ $\cos(53.6°) = \frac{w}{\sqrt{125}}$ ☐ $\tan(90 - x) = \frac{9}{w}$		**Recommended Scoring** **(30 Total Points)**	Consider giving students 6 points for each correct choice and deducting 10 points for each incorrect choice. Students who select all choices should receive 0 points.
Question	**4a**	**Benchmark(s)**	MA.912.T.1.2
$\sqrt{37}$		**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Students may show their work on the coordinate grid. Consider partial credit for minor errors.

Question	4b	Benchmark(s)	MA.912.T.1.2
	$\sqrt{74}$	**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Students may show their work on the coordinate grid. Consider partial credit for minor errors.
Question	**5a**	**Benchmark(s)**	MA.912.T.1.2
	10.4	**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**5b**	**Benchmark(s)**	MA.912.T.1.2
	1.5	**Recommended Scoring** **(7 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	5c	Benchmark(s)	MA.912.T.1.2
5.8		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**6**	**Benchmark(s)**	MA.912.T.1.2
199.695 feet		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.